MobileTransfer: Cross-Dataset Transfer Evaluation for Mobile-Money Fraud Detection

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***Abstract*—Mobile-money fraud detection is often evaluated on one public synthetic dataset at a time, even though practical fraud models may need to move across platforms, simulators, or deployment periods with different class priors and fraud semantics. This short paper introduces *MobileTransfer*, a compact cross-dataset benchmark for PaySim and MoMTSim fraud-detection extracts. The benchmark harmonizes both datasets into a shared tabular schema, uses only raw shared transaction fields, and compares in-domain learning, zero-shot cross-dataset transfer, target-only few-shot learning, and source-pretrained few-shot fine-tuning. Our results show that schema similarity is not sufficient for reliable transfer. PaySim and MoMTSim share nearly identical model inputs after preprocessing, yet their label priors and transaction-type fraud patterns differ sharply. XGBoost transfers reasonably from PaySim to MoMTSim but fails in the reverse direction under precision-recall evaluation, suggesting that severe class-prior shift and dataset-specific fraud behavior both affect transfer. Source-pretrained MLP fine-tuning is not uniformly beneficial: it can be competitive in the PaySim-to-MoMTSim direction, but it provides no meaningful gain for MoMTSim-to-PaySim transfer. These findings position cross-dataset evaluation as an underexplored benchmark need for mobile-money fraud research.**

***Index Terms*—mobile money, fraud detection, transfer learning, PaySim, MoMTSim, few-shot learning, dataset shift**

I. Introduction

Mobile-money systems support deposits, withdrawals, transfers, payments, and other financial activity through mobile devices. Their wide use has increased access to digital finance, but it has also created opportunities for fraudulent transfers, cash-out abuse, and account misuse. Because real mobile-money fraud data are difficult to release, public synthetic datasets have become important for reproducible research.

PaySim is a widely used synthetic mobile-money simulator for fraud-detection studies [1]. MoMTSim is a more recent multi-agent mobile-money transaction simulator calibrated against mobile-money ecosystem behavior [2], with a labeled synthetic dataset released for public use [3]. Prior work has reported strong single-dataset results on PaySim and related mobile-payment data using boosted trees, ensemble methods, and neural models [4]–[6]. Recent MoMTSim work also reports strong classifier performance, with XGBoost performing well on rich mobile-money transaction data [7]. Separately, transfer learning has been studied for financial fraud detection, especially in credit-card settings [8], [9], and benchmark design for biased tabular fraud data has received growing attention [10]. However, cross-dataset transfer for mobile-money fraud detection remains less explored.

In operational mobile-money systems, obtaining substantial real-world labeled fraud data is difficult because of privacy, security, and institutional constraints. Synthetic transaction data are therefore attractive not only for academic benchmarking, but also as a possible starting point for fraud-detection models before deployment data are available. If models trained on one transaction dataset generalize reliably to another, simulator-trained or pretrained fraud detectors could be adapted more confidently to new financial institutions with limited labels. Conversely, if transfer fails even between schema-compatible synthetic datasets, then synthetic-to-real transfer claims should be treated cautiously.

This motivates a cross-dataset stress test: `if a model is trained on one synthetic mobile-money fraud dataset, does it remain useful on another?` This is not simply a schema-matching problem. The PaySim and MoMTSim extracts can be mapped to almost the same tabular fields, including step, transaction type, amount, balances, and a binary fraud label. However, similar columns do not guarantee similar learning behavior because class priors, transaction-type distributions, simulator rules, and fraud-label semantics may differ.

This paper therefore makes a deliberately modest contribution. We do not claim a new single-dataset state of the art. Instead, we propose *MobileTransfer*, a benchmark for testing whether cross-dataset transfer is reliable between PaySim and MoMTSim. The contributions of this work are summarized as follows:

- We propose a harmonized PaySim–MoMTSim transaction representation using only raw shared fields, which is designed to be extendable to other mobile-money transaction datasets with similar tabular structure.
- We propose a cross-dataset benchmark that compares XGBoost, MLP zero-shot transfer, target-only few-shot learning, and source-pretrained MLP fine-tuning. The benchmark is implemented in Python and PyTorch, and the code will be released on GitHub upon acceptance to support future research.
- To our knowledge, this is the first study to show that even when PaySim and MoMTSim share similar tabular field names and model inputs, a model trained on one dataset

can suffer a substantial performance drop when directly tested on the other. We further show that few-shot fine-tuning can help the model adapt to a new target dataset under some transfer directions.

The remainder of this paper is organized as follows. Section II presents the MobileTransfer workflow, including schema harmonization, preprocessing, and transfer settings. Section III reports the experimental evaluation on PaySim and MoMTSim datasets. Section IV discusses the implications and limitations of the observed asymmetric transfer behavior, and Section V concludes the paper.

II. MobileTransfer: Cross-Dataset Method

A. Problem Setting

Let $\mathcal{D}_s = (x_i^s, y_i^s)$ be a labeled source dataset and $\mathcal{D}_t = (x_i^t, y_i^t)$ be a labeled target dataset, where $y \in 0, 1$ indicates legitimate or fraudulent behavior. In cross-dataset transfer, a model trained on $\mathcal{D}_s$ is evaluated on a held-out test split from $\mathcal{D}_t$. In few-shot transfer, the model may additionally use a small balanced target support set containing k fraud and k legitimate examples from $\mathcal{D}_t$. This setting differs from ordinary in-domain evaluation. The target test set is never used for fine-tuning or threshold selection. The main question is whether learned fraud patterns transfer across synthetic mobile-money data sources that appear schema-compatible.

B. Schema and Preprocessing

Table I shows how raw PaySim and MoMTSim columns are aligned into a shared schema. Raw account identifiers are retained only for inspection and duplicate checks because simulator-specific IDs have no stable cross-dataset meaning.

The final model input consists of `step`, `type`, `amount`, `oldbalanceOrg`, `newbalanceOrig`, `oldbalanceDest`, and `newbalanceDest`. For XGBoost, numeric fields are passed through directly and transaction type is one-hot encoded. For the neural model, numeric fields are standardized after fitting the scaler on the corresponding training data, and transaction type is one-hot encoded using the common type order. The neural output is a binary fraud probability.

C. Models

XGBoost is included as a strong tabular baseline because boosted trees are commonly competitive on PaySim and MoMTSim-style fraud data [4], [7]. The neural model is a multilayer perceptron (MLP) with hidden dimensions $[128, 64, 64, 32]$ and dropout 0.20. The MLP uses binary cross-entropy with positive-class weighting to handle class imbalance.

For source-pretrained fine-tuning, the MLP is first trained on the source support pool and then updated on a balanced target support set. For target-only few-shot learning, the same MLP architecture is trained from scratch on the identical target support examples. This comparison isolates whether source pretraining improves target adaptation beyond simply seeing a small balanced target set.

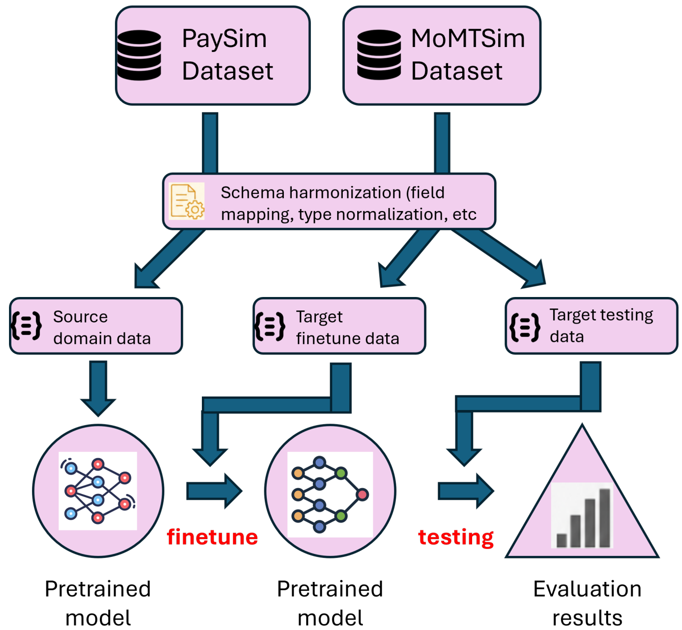


Fig. 1: MobileTransfer workflow. Raw PaySim and MoMTSim CSV files are first harmonized into a shared schema. A source model is pretrained on the source support pool, then adapted using a balanced target support set containing k fraud and k legitimate examples. All methods are evaluated on the same held-out target test set.

III. Experimental Evaluation

A. Datasets and Protocol

The experiment uses Kaggle-hosted fraud-detection extracts derived from PaySim and MoMTSim [11], [12]. Table II summarizes the loaded data. The two extracts have the same number of usable transaction types after harmonization, but their fraud rates differ by more than two orders of magnitude. PaySim contains 8,213 fraudulent transactions out of 6.36 million rows, while MoMTSim contains 2.23 million fraudulent transactions out of 4.23 million rows.

The transaction-type fraud patterns are also different. In PaySim, fraud appears in both `TRANSFER` and `WITHDRAWAL`; in MoMTSim, the fraud labels in this extract are concentrated in `TRANSFER`. This matters because a model trained on MoMTSim may learn a much stronger association between transaction type and fraud than is appropriate for PaySim.

Each dataset is split into an 80% support pool and a 20% held-out test set using stratified random splitting. The support pool is used for source training and for few-shot target support sampling; the test set is reserved for final evaluation. For few-shot experiments, $k \in \{5, 10, 50, 100\}$ fraud examples and k legitimate examples are sampled from the target support pool for each of three random seeds. We exclude $k = 1$ from the main paper because the support set is too small for stable adaptation and thresholded diagnostics. Baseline XGBoost and MLP models use a fixed threshold of 0.5 only for uncalibrated thresholded diagnostics. Few-shot MLP models select the F1-

TABLE I: Field harmonization between PaySim and MoMTSim fraud-detection extracts.

Canonical field	PaySim field	MoMTSim field	Role in benchmark
`step`	`step`	`step`	Numeric model feature
`type`	`type`	`transactionType`	Normalized transaction type; one-hot encoded
`amount`	`amount`	`amount`	Numeric model feature
`nameOrig`	`nameOrig`	`initiator`	Identifier retained for diagnostics; excluded from model input
`oldbalanceOrg`	`oldbalanceOrg`	`oldBalInitiator`	Numeric model feature; sender balance before transaction
`newbalanceOrig`	`newbalanceOrig`	`newBalInitiator`	Numeric model feature; sender balance after transaction
`nameDest`	`nameDest`	`recipient`	Identifier retained for diagnostics; excluded from model input
`oldbalanceDest`	`oldbalanceDest`	`oldBalRecipient`	Numeric model feature; receiver balance before transaction
`newbalanceDest`	`newbalanceDest`	`newBalRecipient`	Numeric model feature; receiver balance after transaction
`isFraud`	`isFraud`	`isFraud`	Binary target label
`isFlaggedFraud`	`isFlaggedFraud`	Not provided; set to 0	Retained for schema compatibility; excluded from main model input

TABLE II: Dataset summary after loading and harmonization.

Dataset	Rows	Frauds	Fraud rate
PaySim	6,362,620	8,213	0.129%
MoMTSim	4,225,958	2,233,118	52.843%

maximizing threshold on the target support set for thresholded metrics, while PR-AUC remains threshold independent.

B. Metrics

Because PaySim is extremely imbalanced, accuracy alone is not informative. We report precision-recall AUC (PR-AUC), F1, and Matthews correlation coefficient (MCC). PR-AUC is the primary ranking metric for rare fraud detection, while MCC summarizes thresholded performance under imbalance.

C. In-Domain and Zero-Shot Results

Table III reports in-domain and zero-shot transfer results. In-domain performance is strong for both datasets, especially for XGBoost. However, direct cross-dataset zero-shot testing produces clear PR-AUC drops relative to the corresponding in-domain baselines. For XGBoost, PR-AUC drops from 0.9969 in-domain on MoMTSim to 0.9644 when trained on PaySim and tested on MoMTSim, and drops more sharply from 0.9609 in-domain on PaySim to 0.0043 when trained on MoMTSim and tested on PaySim. The MLP shows the same directional pattern. These results support the central claim that schema similarity alone does not guarantee zero-shot generalization. The MoMTSim-to-PaySim failure should be interpreted cautiously: it likely reflects severe class-prior shift as well as differences in fraud-type distributions and simulator behavior, rather than feature mismatch alone.

D. Few-Shot Fine-Tuning Results

Table IV compares target-only few-shot MLP training with source-pretrained MLP fine-tuning. Values are mean PR-AUC across three random seeds. The results again show asymmetry. In the PaySim-to-MoMTSim direction, source-pretrained fine-tuning improves adaptation at some shot counts: at $k = 100$, fine-tuning reaches PR-AUC 0.9861 compared with 0.9689. At $k = 5$, $k = 10$, and $k = 50$, target-only training is higher

TABLE III: In-domain and zero-shot results on held-out test sets. PR-AUC is the primary metric; F1 and MCC are computed at the uncalibrated threshold 0.5.

Model	Train $\rightarrow$ Test	PR-AUC	F1@0.5	MCC@0.5
XGB	PaySim $\rightarrow$ PaySim	0.9609	0.3836	0.4854
XGB	MoMTSim $\rightarrow$ MoMTSim	0.9969	0.9701	0.9390
XGB	PaySim $\rightarrow$ MoMTSim	0.9644	0.4444	0.3978
XGB	MoMTSim $\rightarrow$ PaySim	0.0043	0.0153	0.0541
MLP	PaySim $\rightarrow$ PaySim	0.8331	0.1000	0.2260
MLP	MoMTSim $\rightarrow$ MoMTSim	0.9965	0.9691	0.9366
MLP	PaySim $\rightarrow$ MoMTSim	0.7950	0.3440	0.1958
MLP	MoMTSim $\rightarrow$ PaySim	0.0030	0.0084	0.0186

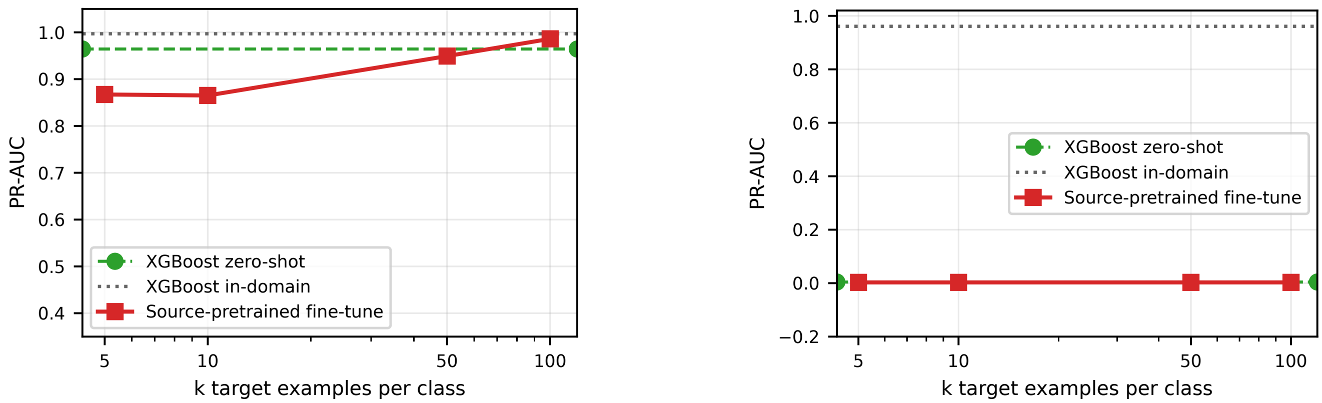
TABLE IV: Few-shot PR-AUC across tested shot counts. Each support set contains k fraud and k legitimate target examples.

Target	k	Target-only	Source-pretrained
MoMTSim	5	0.8941	0.8672
MoMTSim	10	0.9140	0.8651
MoMTSim	50	0.9546	0.9493
MoMTSim	100	0.9689	0.9861
PaySim	5	0.0258	0.0029
PaySim	10	0.0152	0.0029
PaySim	50	0.0729	0.0029
PaySim	100	0.0672	0.0029

on average, although the two methods are close at $k = 50$. In the reverse MoMTSim-to-PaySim direction, source-pretrained fine-tuning provides no meaningful benefit and remains near PR-AUC 0.0029 across all tested support sizes.

IV. Discussion and Future Work

The experiments show that cross-dataset mobile-money fraud detection can fail even when two datasets have nearly identical tabular fields. The clearest case is the MoMTSim-to-PaySim direction: MoMTSim has a much higher fraud rate and fraud is concentrated in `TRANSFER`, while PaySim fraud is extremely rare and appears in both `TRANSFER` and `WITHDRAWAL`. A pretrained neural model can therefore transfer an inappropriate decision bias into the PaySim target domain. This negative result shows why single-dataset performance is insufficient for evaluating mobile-money fraud models and why balanced few-shot support sets are needed when target fraud is rare. At the same time, the PaySim-to-MoMTSim direction shows that source pretraining can



(a) Target: MoMTSim, source: PaySim.

(b) Target: PaySim, source: MoMTSim.

Fig. 2: Few-shot adaptation is asymmetric across PaySim and MoMTSim. PaySim-to-MoMTSim fine-tuning improves PR-AUC at $k = 100$, while MoMTSim-to-PaySim fine-tuning provides no meaningful gain and exhibits negative transfer for all k values.

improve few-shot adaptation when the source bias is less harmful to the target. Overall, direct model reuse is unlikely to be sufficient for new mobile-money environments; adaptation and generalization techniques are needed, but naive fine-tuning may not be enough.

Several limitations remain. Both datasets are synthetic, so cross-dataset success does not imply deployment readiness on real mobile-money data. In addition, thresholds selected on balanced support sets may not reflect natural target prevalence, especially for PaySim. Future work should evaluate temporal splits, anonymized real-world datasets where available, stronger tabular neural models, representation learning, calibration-aware thresholding, and triplet or contrastive encoders.

V. Conclusion

This paper introduced MobileTransfer, a compact benchmark for cross-dataset mobile-money fraud detection. The results show that schema-compatible PaySim and MoMTSim extracts can still produce highly asymmetric transfer behavior: fine-tuning can help in the PaySim-to-MoMTSim direction, while MoMTSim-to-PaySim shows negative transfer. This supports the need to evaluate fraud models beyond single-dataset performance and motivates future work on stronger tabular transfer models, temporal splits, and calibration-aware adaptation.

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